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Information Retrieval
Midterm
2025-10-15

Question 1 (4 points)

Given the following text corpus and a user query. Your task is to answer the following questions based on your understanding of semantic similarity and the vector space model.

Sentence No	Sentence Summary	Cosine Similarity
1	The moon affects the tides on Earth.	0.54
2	Gorillas live in forests...	0.21
3	Math is a universal language..	0.33
4	Space exploration helps us under- stand...	0.92
5	Bananas are rich in potassium...	0.18
6	Astronauts train for years...	0.87

UserQuery: *"Information about the planet or outer space".*

1.1 Which of the following best defines the term semantics in the context of information retrieval?

a. The frequency of words in a sentence.

b. The order of words in a sentence.

c. The grammatical structure of text.

d. The meaning conveyed by the words and sentences

e. I do not know

1.2 What is the main purpose of converting sentences and queries into vectors?

a. To count the number of tokens in the text.

b. To store the text in compressed form.

c. To allow comparison of meaning in a numerical space.

d. To convert text to lowercase.

e. I do not know

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1.3 Based on the similarity scores, which two sentences are most relevant to the query?

a. Sentences 1 and 3

b. Sentences 2 and 5

c. Sentences 4 and 6

d. Sentences 3 and 6

e. I do not know

1.4 If sentence 2 (about gorillas) is ranked as the most relevant, what is the most likely explanation?

a. The query mentions forests.

b. The model failed to capture semantic meaning correctly.

c. The sentence is longer than others.

d. Cosine distance was used instead of similarity.

e. I do not know

Question 2 (1 point)

What is a key consequence of performing PCA on uncentered data?

a. The covariance matrix becomes singular and cannot be decomposed.

b. The first principal component often points toward the mean of the data, rather than the direction of true maximum variance.

c. The eigenvalues of the covariance matrix are reduced to zero.

d. The data becomes overly standardized, losing important patterns.

e. I do not know

Question 3 (1 point)

Why is Singular Value Decomposition (SVD) not as limited as Principal Component Analysis (PCA)?

a. PCA can only be applied to non-square matrices, while SVD can be applied to square ones.

b. SVD can be applied to any real (or even complex) matrix, while PCA requires the data to be mean-centered and based on a covariance matrix.

c. PCA can only work with sparse matrices, while SVD cannot.

d. SVD always gives the same results as PCA but is computationally slower.

e. I do not know

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Question 4 (1 point)

Which statement best describes the Document Object Model (DOM) in web pages?

a. The DOM is a programming language for building HTML structures dynamically.

b. The DOM is a tree-like representation of a web page's elements that allows scripts to access and modify its structure and content.

c. The DOM is a static snapshot of an HTML file that cannot change once loaded.

d. The DOM is a database storing all website resources for the browser.

e. I do not know

Question 5 (1 point)

In the context of representing text as high-dimensional vectors in a vector space, how does increasing the dimensionality of the word or document embeddings theoretically affect the model's ability to capture semantic meaning and the accuracy of similarity calculations?

a. Increasing dimensionality always and monotonically improves the accuracy of semantic similarity scores, as more dimensions allow for a more precise and nuanced representation of meaning.

b. Increasing dimensionality initially improves accuracy by allowing the model to capture more nuanced semantic features, but beyond an optimal point, it can lead to overfitting, increased noise, and a degradation of meaningful similarity due to the "curse of dimensionality."

c. The dimensionality of the vector has no significant impact on the accuracy of similarity scores; only the quality and size of the training data matter.

d. A higher-dimensional vector space inherently reduces the accuracy of similarity scores because the increased sparsity of data points makes all comparisons equally distant.

e. I do not know

Question 6 (1 point)

What is the main data structure used in modern information retrieval systems for mapping terms to document occurrences?

a. Term-document matrix

b. Linked list

c. Inverted index

d. Binary search tree

e. I do not know

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Question 7 (1 point)

What is the purpose of stop word removal during indexing?

a. Reduce query time

b. Increase retrieval precision

c. Reduce index size

d. Prevent duplicate documents

e. I do not know

Question 8 (1 point)

What does the precision metric measure in IR evaluation?

a. Fraction of relevant documents in the retrieved set

b. Fraction of retrieved documents that are relevant

c. Fraction of non-relevant documents in the entire collection

d. Time taken to process a query

e. I do not know

Question 9 (1 point)

Heaps' Law is used to estimate what property of collections?

a. Document length

b. Growth of vocabulary

c. Number of queries per second

d. Size of postings lists

e. I do not know

Question 10 (1 point)

What is the main advantage of blocked dictionary storage in IR systems?

a. Faster searches

b. Smaller dictionary size

c. Simpler code

d. More accurate retrieval

e. I do not know

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Question 11 (1 point)

What is a postings list?

a. A list of terms in a document

b. A list of documents containing a given term

c. A summary of document topics

d. A sequence of user queries

e. I do not know

Question 12 (1 point)

What does Zipf's Law describe with respect to words in large collections?

a. Vocabulary size

b. Frequency distribution of terms

c. Rate of disk accesses

d. Precision-recall tradeoff

e. I do not know

Question 13 (1 point)

Which of the following is NOT a typical reason for compressing IR indices?

a. Save disk space

b. Increase query accuracy

c. Speed up data transfer

d. Enable more caching

e. I do not know

Question 14 (1 point)

In large-scale IR index construction, what role does MapReduce play?

a. Compressing terms

b. Distributed processing of indexing tasks

c. Ranking documents by relevance

d. Normalizing token types

e. I do not know

Question 15 (1 point)

What is dynamic indexing?

a. Indexing that runs in memory only

b. Indexing that updates to reflect additions and deletions

c. Indexing only when prompted by a user

d. Indexing distributed across many computers

e. I do not know

Question 16 (1 point)

What is a key characteristic of a postings list for a frequent term after gap encoding?

a. Mostly zero gaps

b. Mostly gaps of length one

c. Large variable gaps

d. No gaps encoded

e. I do not know

Question 17 (1 point)

In evaluation of ranked retrieval, what does 'Mean Average Precision' (mAP) measure?

a. Average query processing time

b. Mean of precision scores over queries

c. Rank of the most relevant document

d. Number of relevant results shown at the top

e. I do not know

Question 18 (1 point)

If a postings list for 'home' is [1, 2, 4], what does this mean?

a. 'home' appears three times in one document

b. Documents 1, 2, and 4 all contain 'home'

c. 'home' is a stop word

d. The inverted index is corrupted

e. I do not know

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Question 19 (5 points)

Write a function that takes a list of words and constructs an inverted k-gram index. A k-gram is a contiguous substring of length k from a word. The function should return a dictionary where keys are k-grams and values are lists of words that contain those k-grams.

Function Signature (in Python):

```
def build_k_gram_index(words, k=2):
```

Requirements:

- Extract all possible k-grams from each word using a sliding window approach
- For each k-gram, maintain a list of all words that contain it
- Handle words shorter than k by either skipping them or including them with padding (your choice, but document it)

Example Input:

words = ["apple", "banana", "apply", "cat", "bat", "data"]

k = 2

Your task:

- Write the function implementation
- Show the output dictionary for the given example
- Briefly explain your approach

Expected Output Structure:

The function should return a dictionary like:

```
{
  "ap": ["apple", "apply"],
  "pp": ["apple", "apply"],
  "pl": ["apple", "apply"],
  "le": ["apple"],
  "ba": ["banana", "bat"],
  "an": ["banana"],
  "na": ["banana"],
```

```
  "ca": ["cat"],
  "at": ["cat", "bat"],
  "bt": ["bat"],
  "da": ["data"],
  "ta": ["data"]
}
```

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Question 20 (5 points)

Write Python code that takes a list of documents (each document is a string) and constructs an inverted index.

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Question 21 (5 points)

Given an inverted index and a list of query terms, write a function that returns the set of documents containing all the query terms.